Understanding the Structural Differences between Spherical and Rod-Shaped Human Insulin Nanoparticles Produced by Supercritical Fluids Precipitation

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In this study, the thermal denaturation mechanism and secondary structures of two types of human insulin nanoparticles produced by a process of solution-enhanced dispersion by supercritical fluids using dimethyl sulfoxide (DMSO) and ethanol (EtOH) solutions of insulin are investigated using spectroscopic approaches and molecular dynamics calculations. First, the temperature-dependent IR spectra of spherical and rod-shaped insulin nanoparticles prepared from DMSO and EtOH solution, respectively, are analyzed using principal component analysis (PCA) and 2D correlation spectroscopy to obtain a deeper understanding of the molecular structures and thermal behavior of the two insulin particle shapes. All-atom molecular dynamics (AAMD) calculations are performed to investigate the influence of the solvent molecules on the production of the insulin nanoparticles and to elucidate the geometric differences between the two types of nanoparticles. The results of the PCA, the 2D correlation spectroscopic analysis, and the AAMD calculations clearly reveal that the thermal denaturation mechanisms and the degrees of hydrogen bonding in the spherical and rod-shaped insulin nanoparticles are different. The polarity of the solvent might not alter the structure or function of the insulin produced, but the solvent polarity does influence the synthesis of different shapes of insulin nanoparticles.

1. Introduction

Insulin is used to treat advanced-stage diabetes and is administered to patients by injection, orally, or by pulmonary inhalation. In the case of injected insulin, however, patients can experience discomfort and might not use antiseptic techniques. Furthermore, drug delivery through the skin is inefficient because proteins and/or peptides, such as insulin, have large molecular weights and tertiary structures.[1–6] Oral administration also has limitations because of acidic and enzymatic degradation of insulin in the gastrointestinal tract and the low permeability and instability of the formulations under these conditions.[7,8] Therefore, the pulmonary delivery of insulin has been investigated in clinical studies. For a pulmonary delivery system, it is important for the particles to be less than 5 μm in size, to have a narrow particle-size distribution and to possess biological activity. Insulin particles smaller than 5 μm can be used for delivery through the lungs and for controlled drug release.[1–5]

The solution-enhanced dispersion by supercritical fluids (SEDS) process is an interesting and powerful technique for producing protein nanoparticles with a narrow particle size distribution.[4–6,8–10] Surprisingly, protein nanoparticles fabricated with supercritical fluids retain their biological activity and native structure upon redissolution in aqueous solution.[1,4–6,8–10] Pyo reported that two shapes of insulin nanoparticles can be produced using the SEDS process from dimethyl sulfoxide (DMSO) and ethanol (EtOH) solutions without the use of additives.[5] DMSO and EtOH were used to assist the supercritical carbon dioxide extraction of water from the aqueous protein solution. The difference between the interfacial tensions of the DMSO and EtOH solutions were hypothesized to result in the precipitation of insulin nanoparticles with different shapes. However, the structural differences between the two types of insulin nanoparticles have not yet been investigated.

Infrared (IR) spectroscopy is a valuable tool for investigating changes in the secondary structures of proteins, particularly based on the amide I region.[11–14] The amide I band is the most prominent band in the IR spectrum of a protein, and a well-defined relationship exists between the frequency of the band and the secondary structure of the protein. However, the extensive overlap of spectral features occasionally limits the utility of IR spectroscopy for the elucidation the structural changes in proteins. To overcome the shortcomings of conventional spectral analysis, 2D correlation spectroscopy has been applied to IR spectra.

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2D correlation spectroscopy is a powerful technique for interpreting spectra obtained under external perturbations, as previously demonstrated, and the details of the technique are described in those reports.[15,16] The primary advantage of 2D correlation spectroscopy is that it can be used to probe the specific sequence of spectral intensity changes that occurs during the perturbation-controlled measurement, which is barely detectable in conventional spectra. Therefore, 2D IR correlation spectroscopy has been extensively used in fundamental protein research because it allows the evolution of protein structures to be explored.[17–23]

In this study, to elucidate the molecular structures and thermal behavior of insulin nanoparticles of two different shapes, the temperature-dependent IR spectra of the particles were analyzed using 2D correlation spectroscopy. A principal component analysis (PCA) was also performed to monitor the evolution of the spectral changes of the two shapes of insulin with increasing temperature. PCA provides a precise mathematical analysis of the changes in the sample and in the variable vectors, and is a well-established analytical technique for spectroscopic studies.[24–26] 2D correlation spectroscopy combined with PCA enables improved characterization of the structural changes in biomolecules at a detailed molecular level.[17,23]

The all-atom molecular dynamics (AAMD) simulation method is a powerful tool that enables the examination of complex systems and processes at the atomic level. Due to significant advances in computing power, AAMD simulations have been widely used to investigate the structural and dynamic properties of condensed-phase systems.[27,28] The computational results from AAMD simulations primarily depend on the force field, and several accurate and reliable force fields have been developed for proteins and organic molecules.[29,30] In this study, we performed a series of AAMD simulations for an insulin monomer solvated in water, DMSO/water, and EtOH/water to investigate the influence of solvent molecules on the formation of insulin nanoparticles and to examine the origin of the geometric difference between the two shapes of particles.

2. Results and Discussion

Spherical and rod-shaped insulin nanoparticles were produced using the SEDS technique with DMSO and EtOH as the solvents, respectively, as shown in Figure 1, which presents field-emission scanning electron microscopy (FE-SEM) images of the synthesized nanoparticles.

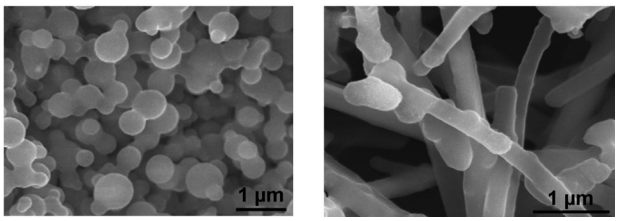

Figure 1. FE-SEM images of a) spherical and b) rod-shaped nanoparticles produced using the SEDS method with DMSO and EtOH solvents, respectively.

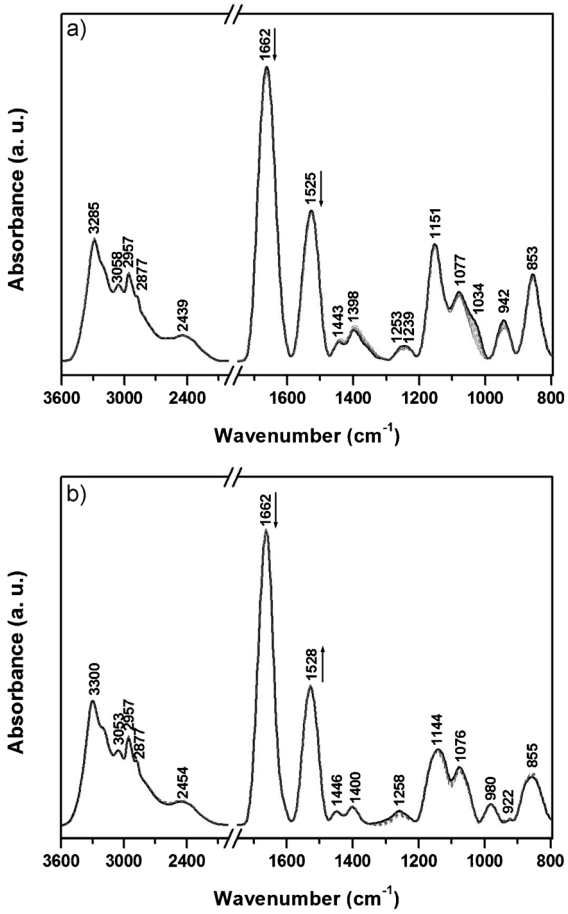

Figure 2. Temperature-dependent IR spectra in the 800–3600 cm^{-1} region for the a) spherical and b) rod-shaped insulin nanoparticles. The arrows indicate the direction of changes in the band intensity with increasing temperature.

The temperature-dependent IR spectra of these two shapes of insulin nanoparticles are shown in Figure 2. The changes in the IR spectra with increasing temperature were similar for the spherical and rod-shaped nanoparticles in the amide A/B (3600–2750 cm^{-1}) and amide I (1800–1580 cm^{-1}) regions. The shapes and positions of the bands in the amide A/B and amide I regions of the spectra of both types of insulin at room temperature (25 °C) resembled those found in the previously reported spectrum of a dry film of native insulin.[28] This result indicates that the conformation of the insulin in the spherical and rod-shaped nanoparticles produced using SEDS was similar to that in the native state. As indicated by the spectra collected at room temperature shown in Figure 2, one amide A band at 3285 cm^{-1} in the spherical nanoparticles was observed, whereas a band appeared at 3300 cm^{-1} in the spectrum of the rod-shaped nanoparticles. The amide A band due to N–H stretching is sensitive to the degree of hydrogen bonding of the main-chain NH groups.[28] Therefore, the degree of hydrogen bonding in the main-chain NH groups of spherical nanoparticles was greater than that in the rod-shaped nanoparticles, as demonstrated by the lower frequency amide A band for spherical insulin nanoparticles relative to that ob-

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served with the rod-shaped nanoparticles. Noticeable changes in four bands, at 1443, 1398, 1034, and 942 cm^{-1}, were observed in the temperature-dependent IR spectra of the spherical nanoparticles (Figure 2a). A band at 1034 cm^{-1} related to the C–O stretching in the side-chains of serine residues appeared only in the IR spectra of the spherical nanoparticles, and this band decreased with increasing temperature. This result indicates that serine might predominantly contribute to the temperature-dependent changes in the secondary structure of spherical insulin nanoparticles. In this study, we only focused on the amide I region (1750–1585 cm^{-1}) of the two shapes of insulin nanoparticles to better understand the secondary structure changes that occur during thermal denaturation.

Based on data in the literature,[1,18,28,31–34] the band assignments for the IR spectra of the spherical and rod-shaped insulin nanoparticles are summarized in Table 1. However, the amide I regions of proteins, which exhibit highly overlapped bands, are difficult to interpret accurately with respect to structural changes. To better understand the variations in the secondary structures of the particles as a function of increasing temperature, PCA and 2D correlation analysis were applied to the temperature-dependent IR spectra of the spherical and rod-shaped insulin nanoparticles.

The plots of scores and loading vectors of the first three principal components (PCs) are shown in Figures 3 and 4, which were obtained from the amide I region in the temperature-dependent IR spectra of the spherical and rod-shaped insulin nanoparticles, respectively. The data in Figure 3 indicates that the first principal component (PC1) accounts for more than 96.22% of the total absorbance change in the spectra of the spherical nanoparticles, whereas the second principal component (PC2) captures 2.66%, which corresponds to the majority of the remaining variances not described by PC1. For the rod-shaped nanoparticles, PC1 and PC2 accounted for 85.27% and 12.65% of the variance, respectively (Figure 4). The PC1 scores of both the spherical and rod-shaped nanoparticles varied from positive to negative values at approximately 55 °C (Figures 3a and 4a). For the spherical nanoparticles, the loading vector for PC1 at 1672 cm^{-1} had a positive value, whereas those at 1716 and 1603 cm^{-1} had negative values (Figure 3b); for the rod-shaped insulin nanoparticles (Figure 4b), the loading vector for the PC1 value at 1658 cm^{-1} was only positive. This result indicates that thermal denaturation of the spherical nanoparticles occurred at all temperatures tested, whereas thermal denaturation of the rod-shaped nanoparticles occurred at temperatures below 55 °C. Furthermore, the major components of the secondary structure of the spherical nanoparticles were different from those of the rod-shaped nanoparticles; β structures, such as β turns and β sheets, were the major structural components in the spherical nanoparticles, whereas helical structures (e.g. α helix or 3^1 helix) were the major components in the rod-shaped nanoparticles.

Figure 5 and 6 show the 2D IR correlation spectra of the spherical and rod-shaped insulin nanoparticles, respectively. In

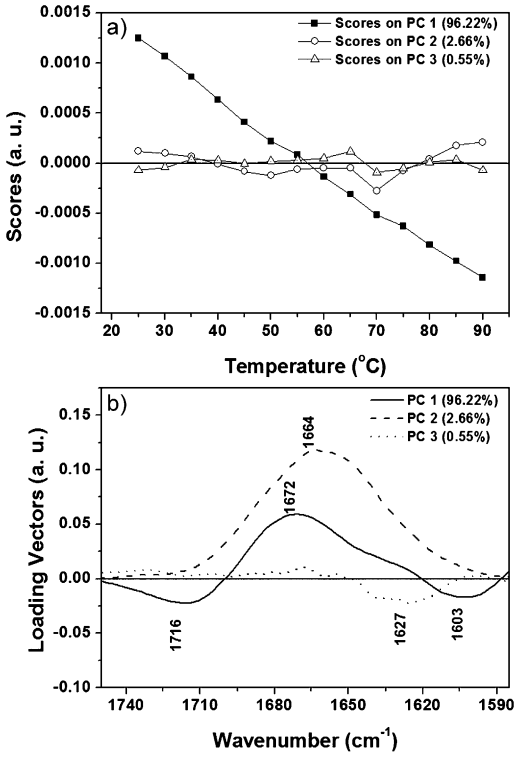

Figure 3. a) Score plots and b) plot of loading vectors for the first three PCs of the temperature-dependent IR spectra of spherical insulin nanoparticles.

Table 1. Band assignments for the spherical and rod-shaped insulin nanoparticles as a function of temperature changes observed in the IR (1D) spectra and the 2D correlation spectra.

	Wavenumber [cm^{-1}]			
	Spherical		Rod shape	
Band assignments	1D spectra	2D spectra	1D spectra	2D spectra
Amide A	3285	NA	3300	NA
Amide B	3058	NA	3053	NA
C–H stretching	2957	NA	2957	NA
C–H stretching	2877	NA	2877	NA
Cys ν(S–H)	2439	NA	2454	NA
Intermolecular β sheet	–	1714	–	–
β sheet	1662	–	1662	1689
β turn	–	1674	–	1672
α helix or 3^{10} helix	–	1664	–	1664
Disordered structure	–	1647	–	1641
β sheet	–	–	–	1630
β sheet	–	1610	–	–
δ(N–H) + vibration from side chain	1525	NA	1528	NA
δ(CH_3)	1443	NA	1446	NA
Gln ν(CN)	1398	NA	1400	NA
Glu δ(COH)	1253	NA	1258	NA
His δ(CH)	1239	NA	–	NA
Glu ν(C–O)	1151	NA	1144	NA
His δ(CH)	1077	NA	1076	NA
Ser ν(C–O)	1034	NA	–	NA
Pro ν(CN) or ν(CC)	942	NA	980	NA
Gly γ(CH_2)	–	NA	922	NA
γ(CH_2)	853	NA	855	NA

NA: 2D correlation analysis was not performed.

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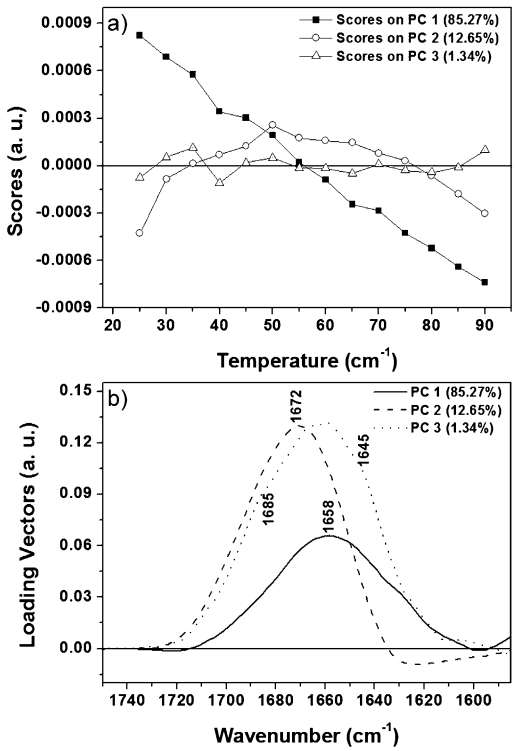

Figure 4. a) Score plots and b) plot of loading vectors for the first three PCs of the temperature-dependent IR spectra of rod-shaped insulin nanoparticles.

a synchronous 2D correlation spectrum, a positive cross peak at (ν_1, ν_2) indicates that the spectral intensities of two bands are either increasing or decreasing together as functions of the external perturbation during the observation interval. In contrast, a negative sign for the cross peaks indicates that one of the spectral intensities is increasing while the other is decreasing. The sequential order of the spectral intensity changes can be determined from the sign of the synchronous and asynchronous cross peaks. If the signs of these cross peaks are the same, the intensity change at ν_1 occurs before that at ν_2. However, if the signs of the synchronous and asynchronous cross peaks differ, the intensity change at ν_1 occurs after that at ν_2.[15,16] As shown in the synchronous 2D IR correlation spectra (Figures 5a and 6a), three bands at 1716, 1670, and 1603 cm^{-1} were observed in the spherical insulin nanoparticles, whereas the bands at 1664 and 1630 cm^{-1} were the most changed in the rod-shaped nanoparticles. The synchronous 2D IR correlation spectra clearly revealed that variations in the β structure with increasing temperature predominated in the spherical nanoparticles, whereas in the rod-shaped nanoparticles, the helical structures were sensitive to temperature. These results are in good agreement with the PCA, revealing that the principle components of the secondary structure that change with increasing temperature were clearly different between the two types of particles. The signs of the cross peaks in the 2D IR correlation spectra of the spherical insulin nanoparticles (Figure 5) suggests that the sequence of changes in the spectral intensities with heating occurred in the following order: 1647 (due to disordered structure) → 1664 (α helix) → 1674 (β turn) → 1714 (intermolecular β sheet) → 1610 cm^{-1} (β sheet).

As shown in Figure 6, the following sequence of changes in the spectral intensities was observed for the rod-shaped particles with increasing temperature: 1630 (due to β sheet) → 1641 (disordered structure) → 1664 (α helix) → 1672 (β turn) → 1689 cm^{-1} (β sheet). From the 2D correlation analysis, it can be concluded that an intermolecular β-sheet structure exists only in the spherical insulin nanoparticles and that the thermal behavior of the two shapes of nanoparticles are completely different.

To thoroughly investigate the conformations of the two types of insulin nanoparticles, we performed AAMD simulations of an insulin monomer solvated in DMSO/water and in EtOH/water. For comparison, AAMD simulations of an insulin monomer solvated in water were also performed. Although the simulations were conducted for an insulin monomer rather than nanoparticles, we believe that such simulations provide insight into the formation and properties of insulin nanoparticles at the atomic level. Figure 7 (see also Figure S1 of the Supporting Information) shows the transient numbers of intramolecular, interstrand, and intrastrand hydrogen bonds for an insulin monomer in all three solvents under the conditions of the SEDS process at 313 K and 100 bar. Although the transient and average numbers of intrastrand hydrogen bonds in the monomer have similar values in the three solvents (Figure 7c and d, Table S1), those of the intramolecular and interstrand hy-

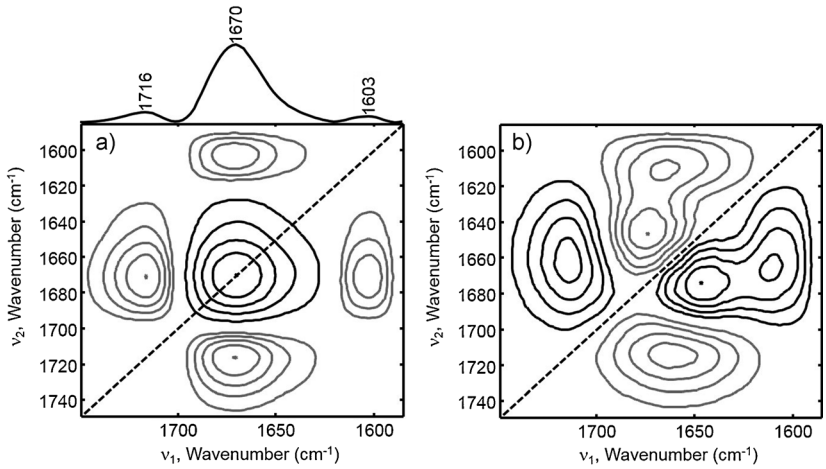

Figure 5. a) Synchronous and b) asynchronous 2D IR correlation spectra obtained from the temperature-dependent changes in spectra of the spherical insulin nanoparticles. The black and gray lines represent positive and negative cross peaks, respectively.

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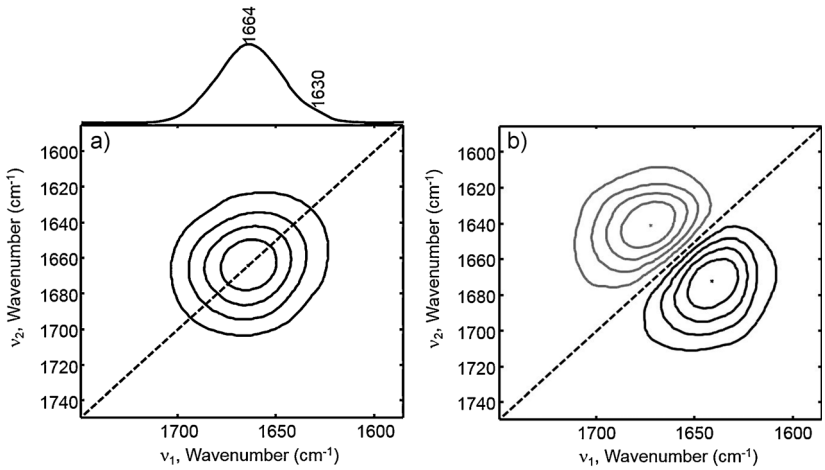

Figure 6. a) Synchronous and b) asynchronous 2D IR correlation spectra obtained from the temperature-dependent changes in spectra of the rod-shaped insulin nanoparticles. The black and gray lines represent positive and negative cross peaks, respectively.

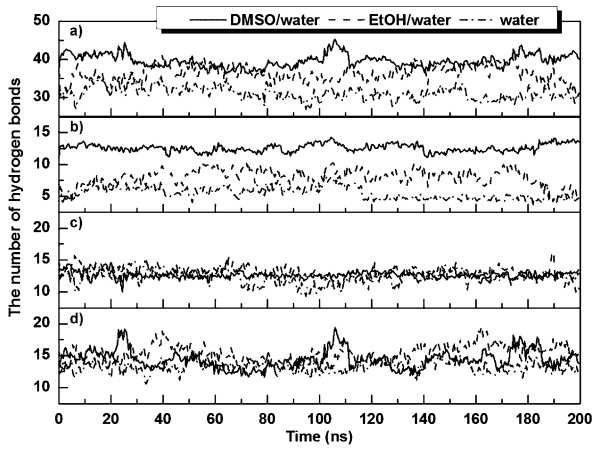

Figure 7. Transient numbers of a) intramolecular, b) interstrand, c) insulin A-chain, and d) B-chain intrastrand hydrogen bonds in an insulin monomer in three solvents at 313 K and 100 bar. The transient numbers were obtained by calculating the average number of hydrogen bonds every 400 ps.

drogen bonds in the monomer in EtOH/water and in water are clearly lower than those found in the monomer in DMSO/water (Figure 7 a and b, Table S1). This result suggests that compared to DMSO, which is a typical polar aprotic solvent, polar protic solvents such as EtOH and water strongly interact with each strand of insulin through hydrogen bonding. These interactions give rise to an increased number of hydrogen bonds between an insulin monomer and solvent molecules in EtOH/water and water solvents (Figure S2), and thus the hydrogen-bond interactions between two individual strands are diminished. We hypothesize that the geometric difference between the two insulin nanoparticles in DMSO/water and in EtOH/water solutions can be attributed to the influence of the solvent on the formation of hydrogen bonds between the two strands of an insulin monomer.

Ramachandran plots for each chain of the insulin monomer in the three solvents are presented in Figure 8 (the sequence of insulin is shown in Figure S3; see also Figure S4). These plots demonstrate that there are significant conformational differences in the non-helical regions.[28] In contrast, the α-helical structure of each strand is well maintained for 200 ns regardless of the solvent used. The persistence of the helical conformation is also confirmed by the average per-residue α-helicity of each chain of the insulin monomer in the three solvents, calculated using the program STRIDE (Figure S5).[35] Indeed, organic solvents such as DMSO or 2,2,2-trifluoroethanol have been reported to enhance the helical conformation of small peptides.[36,37] Although longer AAMD simulations and further calculations are required to assess the helical preference of insulin in organic solvents, the Ramachandran plots imply that the structure and function of insulin might not be significantly altered during the synthesis of insulin nanoparticles in DMSO/water or EtOH/water solutions.

5. Conclusions

We first investigated the thermal denaturation mechanism and secondary structures of two types of human insulin nanoparticles produced using the SEDS process with DMSO and EtOH solutions. The temperature-dependent IR spectra were analyzed using PCA and 2D correlation spectroscopy to elucidate the changes in secondary structure with increasing temperature. AAMD simulations were also performed to compare the molecular structures of each type of insulin nanoparticle. The PCA results and the 2D correlation spectra clearly revealed that the mechanisms of thermal denaturation and the major structural components in the spherical and rod-shaped nanoparticles were different: intermolecular β sheets appeared only in the spherical insulin nanoparticles upon increases in temperature. The data demonstrate that the major structural components of the spherical and rod-shaped nanoparticles are β and helical structures, respectively. From the AAMD simulation, we determined that the polarity of the solvent might not alter the structure or function of insulin but that the solvent polarity does influence the formation of different shapes of insulin nanoparticles.

Although the conformations of insulin in the spherical and rod-shaped nanoparticles synthesized using SEDS were similar, the degree of hydrogen bonding and the thermal behavior of the two nanoparticles were different. Thus, the formation of two different shapes is a result of the different degrees of hydrogen bonding.

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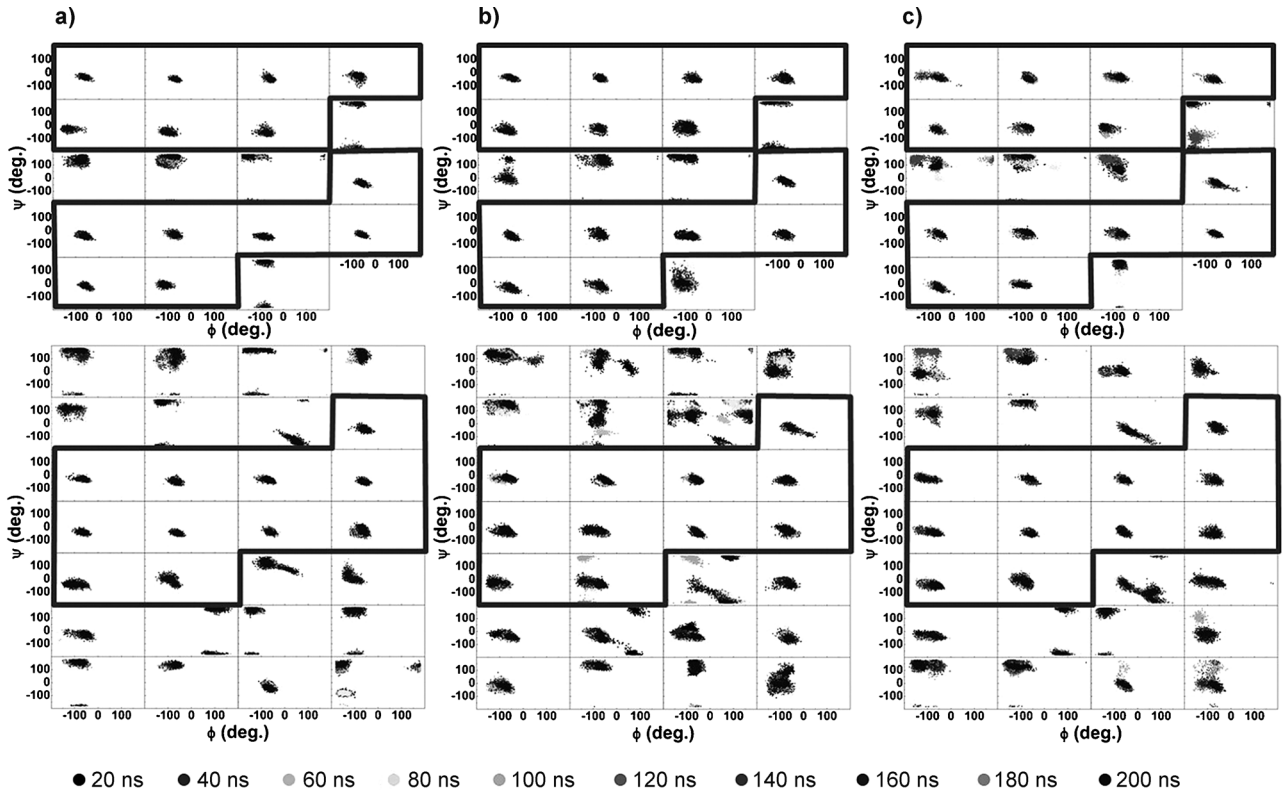

Figure 8. Time evolution of Ramachandran plots for each residue of the insulin monomer at 313 K and 100 bar in a) DMSO/water, b) EtOH/water, and c) water. The black boxes represent the α-helical regions in each strand.

Experimental Section

DMSO (99.9%), EtOH (99.9%), and water (99.9%) were of HPLC grade, purchased from Sigma–Aldrich, and were used as received. The insulin was supplied by Regeron, Inc. (Chuncheon, Korea). CO_2 (99.9%) was supplied by Gold-Star Oxygen Ltd (Chuncheon, Korea). The insulin solution (100 IU mL^{-1}, 1 mL) was mixed with DMSO or EtOH (95%, 50 mL). Spherical and rod-shaped insulin nanoparticles were produced using the SEDS technique with DMSO and EtOH as the solvents, respectively. FE-SEM images of these insulin nanoparticles were recorded on an S-4300 FE-SEM microscope (Hitachi, Tokyo, Japan) operated at 15.0 kV. The nanoparticles were prepared under the following conditions: 40 °C, 100 bar, a CO_2 flow rate of 20 mL min^{-1}, and DMSO or EtOH flow rate of 0.6 mL min^{-1}. The experimental procedure has been described in greater detail elsewhere.[5,6]

The IR spectra were recorded using a DA8 FT spectrometer (Bomem, Hartmann & Braun Quebec, Canada) equipped with a liquid-nitrogen-cooled mercury–cadmium–telluride (MCT) detector at the Pohang Accelerator Laboratory. A thermal diffuse reflectance infrared Fourier transform (DRIFT) accessory was used. For each spectrum, 256 scans were recorded with a spectral resolution of 4 cm^{-1}. For the DRIFT measurements, the spherical and rod-shaped insulin nanoparticles were diluted to 3 wt% in ground KBr. Dynamic spectra were obtained by increasing the temperature from 25 to 90 °C in increments of 5 °C.

Data processing, including baseline correction, smoothing, and area normalization, was performed before the PCA and the 2D correlation analysis. The OriginPro 8.0 program (OriginLab Co., Northampton, MA) was used in the pretreatment for the temperature-dependent IR spectra, including baseline correction, smoothing, and area normalization. Synchronous and asynchronous 2D correlation spectra were obtained using in-house software based on the MATLAB R2010a program (The Mathworks Inc., Natick, MA), in which the black and gray lines in the 2D correlation spectra represent positive and negative cross-peaks, respectively. The PCA was performed using PLS Toolbox 7.0 (Eigenvector Research, Inc., Wenatchee, WA) for MATLAB.

Because AAMD simulations for an entire insulin nanoparticle are computationally demanding, we performed AAMD simulations for a single insulin monomer solvated in three different solvents. The initial coordinates for the heavy atoms of the insulin monomer were obtained from hexameric human insulin at a resolution of 1.64 Å (PDB ID: 4F1G). With the hydrogen atoms properly attached, the insulin monomer was used without further modification. The monomer was then placed at the center of a simulation box in which fully equilibrated DMSO and water, EtOH and water, or water molecules had been prepared. Solvent molecules that overlapped with the inserted insulin monomer were removed to prevent weak van der Waals interactions. The energy of the system was minimized to remove undesired contacts between atoms. The initial dimensions of each simulation box were 60.00×60.00×60.00 Å^3. The number of solvent molecules in each simulation box was determined based on the experimental solvent densities and volume ratios. Finally, three simulation systems were prepared. All of these systems included one insulin monomer. The first system also included 1677 DMSO and 136 water molecules. In the second system, one insulin monomer was solvated in 2013 EtOH and 131 water molecules. The third system consisted of one insulin monomer and 6986 water molecules. The AMBER force field was used for the insulin monomer[29] and the TIP3P water model was used

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for water. The force field for DMSO and EtOH was that reported by Fox and Kollman.[30] The organic solvent models are fully compatible with the AMBER force field, and they have been widely used together with the TIP3P water model to describe EtOH/water or DMSO/water mixtures.[38,39]

The molecular dynamics simulations were performed using the NpT ensemble with periodic boundary conditions in all directions. The temperature and pressure were maintained using the Langevin thermostat and Langevin piston methods, respectively.[40] Electrostatic interactions were calculated using the particle mesh Ewald (PME) method with 64.0×64.0×64.0 grid points for the reciprocal space sum.[41] Dispersion interactions were described using the Lennard–Jones potential with a cutoff distance of 12 Å and a switch distance of 10 Å. For each system, two AAMD simulations were conducted. The first simulation was run for 200 ns at the standard state (298.15 K, 1 bar), followed by the second simulation for 200 ns at 313.15 K and 100 bar, which are similar to the experimental conditions (40.0 °C, 100.0 bar). To emulate the experimental procedure, we ran simulations at 305.15 K and 100 bar from the last configuration of the standard state simulations. For each simulation of 200 ns, an equilibration run was performed for the first 100 ns and then a production run was conducted for the second 100 ns. The time step was 2 fs and the data were collected every 2 ps during each production run. All of the AAMD simulations presented here were performed using the NAMD program, version 2.8.[42]

Acknowledgements

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Keywords: 2D correlation spectroscopy · insulin nanoparticles · molecular dynamics simulation · protein structure · solution-enhanced dispersion by supercritical fluids process

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